Chapter 2: Markets and Price Mechanisms

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Outline

1. Introduction and Motivation
2. Supply, Demand, and Partial Equilibrium
3. Perfect Competition and Price Adjustment
4. Imperfect Competition: Monopoly and Oligopoly
5. Oligopoly Models
6. Wrap-Up and Further Steps

Why Study Markets and Prices?

- **Core of Microeconomics**: Most resource allocation in real economies happens through some market mechanism.
- **Practical Relevance**: Taxes, subsidies, and regulations all operate through market price effects.

Goal of This Chapter

Understand **how prices are determined**, how markets reach *partial equilibrium*, and how **market power** affects outcomes.

Roadmap for Chapter 2

1. **Supply & Demand Analysis**
   - Partial Equilibrium Basics
   - Shifts vs. Movements
   - Examples & Policy Exercises
2. **Perfect Competition**
   - Market Setting, Price-Taking Behavior
   - Short-run vs. Long-run Equilibrium
   - Comparative Statics
3. **Imperfect Competition**
   - Monopoly Model
   - Oligopoly (Cournot, Bertrand, Stackelberg)
   - **Price Discrimination** (Different Degrees)

Supply, Demand, and Partial Equilibrium

Definition (Demand Function)

$Q^D = D(p)$ is the **quantity demanded** by consumers at price p.

Definition (Supply Function)

$Q^S = S(p)$ is the **quantity supplied** by sellers at price p.

Partial Equilibrium Concept

We examine a single market in isolation, taking other prices and incomes as given.

Equilibrium Condition

Market Equilibrium occurs where:

$$Q^D = D(p^*) \quad = \quad Q^S = S(p^*)$$

- p^* is the **equilibrium price**.
- At p^*, supply equals demand.

Example (Example)

If $D(p) = 100 - 2p$ and $S(p) = 10 + 4p$, then set $100 - 2p = 10 + 4p$ to solve for the equilibrium p^*.

Shifts vs. Movements on the Curves

Movement along the demand curve:

- Price changes $\implies$ quantity demanded changes (a move along the same curve).

Shift of the demand curve:

- Income, tastes, or price of related goods changes $\implies$ the entire curve shifts.

Similarly for supply:

- Changes in input costs or technology $\implies$ supply curve shifts.

Comparative Statics: Tax on Sellers

- Suppose a per-unit tax τ is imposed on **producers**.
- Effective supply function shifts up by τ.
- Equilibrium price increases, quantity decreases.

Example (Burden of Tax)

Buyers pay a higher price, but sellers receive lower net price ($p - \tau$). The **tax incidence** *depends on demand and supply elasticities.*

Policy Insight: Government Interventions

- **Price ceilings** (e.g., rent control) can create shortages if set below equilibrium price.
- **Price floors** (e.g., minimum wage) can create surpluses if set above equilibrium price.
- **Quotas** restrict quantity traded, raising prices if demand is strong.

Lesson

Supply-demand analysis clarifies who benefits or suffers from price regulations. This partial equilibrium approach is a starting point for real policy debates.

Perfect Competition: Key Assumptions

- Large number of **identical** firms and many consumers.
- Each firm is a **price-taker**: no single firm can influence market price.
- **Homogeneous** product, free entry/exit in the long run.
- Perfect information about prices, technology, etc.

Why Study It?

Although real markets rarely fit all assumptions, perfect competition sets a useful "benchmark" for efficiency and price determination.

Firm's Decision in Perfect Competition

- Firms take the market price p as given.
- They choose quantity q to maximize profit:

$$\max_{q \geq 0} \pi(q) = p \cdot q - C(q),$$

  where $C(q)$ is the cost function.
- First-order condition (if interior solution):

$$p = MC(q^*).$$

Supply Curve Above AVC

In the short run, a firm continues producing as long as $p \geq \text{AVC}$. If $p < \text{AVC}$, it is cheaper to **shut down** temporarily because the firm cannot cover its variable costs. Hence, the short-run supply curve is the MC curve but **only** for the portion above AVC.

Short-Run Equilibrium

- Sum of **individual firm supplies** $\rightarrow$ market supply.
- Intersection with **market demand** $\rightarrow$ short-run equilibrium price p^*.
- Firms produce q_i^* where $p^* = MC(q_i^*)$ (assuming no shutdown).

Example (Example)

If each competitive firm has supply $S_f(p)$, and there are N identical firms, then total supply $S(p) = N \cdot S_f(p)$.

Producer Surplus in the Short Run

- **Producer Surplus (PS)**: The difference between the market price and the firm's marginal (or variable) cost, summed over the units sold.
- Graphically, area above the supply curve and below the price line.

Remark

PS is not necessarily the same as profit if we include fixed costs. In the short run, fixed costs are sunk, so PS can be seen as revenue above variable costs.

Long-Run Equilibrium in Perfect Competition

- Firms can **enter or exit** the market freely in the long run.
- In equilibrium, **economic profit** must be zero: $p^* = AC$.
- However, **AC** is generally a U-shaped curve, so the price must match its *minimum* point:

$$p^* = \min AC.$$

- **Why the minimum AC?** - If p exceeds $\min AC$, a firm can **adjust its output** to the level where its average cost is minimized, thus making **positive profit**. This attracts new entrants, which drive price back down. - If p is below $\min AC$, existing firms have losses, causing exits and pushing price up.

Conclusion

Only at $p = \min AC$ do all firms earn zero profit and have no incentive to enter or exit.

Real-World Relevance: Perfect Competition

- Commodity markets (e.g., wheat, corn) approximate competition fairly well.
- Tech or brand-differentiated markets seldom are purely competitive.
- Still, the **competitive model** offers intuition: free entry/exit pushes profits down, typically benefiting consumers.

Example: Agricultural Market

- Many small farmers producing nearly identical crops (like wheat).
- Each farmer is too small to affect the global price.
- If prices rise, more farmers plant wheat next season $\rightarrow$ supply expands, price falls back toward equilibrium.

Lesson

Price-taking behavior with many producers is the hallmark of perfect competition, though real complications (weather, subsidies) exist.

Why Study Imperfect Competition?

- Many real markets have a **dominant** firm or a few major players (e.g., Big Tech, telecom).
- Prices can be manipulated by firms with market power.
- Government policy (antitrust, regulation) tries to control monopolistic or collusive behaviors.

Goal

Analyze how **market power** changes pricing/output, compare it to the competitive benchmark, and see welfare implications.

Monopoly: Basic Setup

- Single seller faces market demand $Q = D(p)$ or equivalently $p = p(Q)$.
- **Maximize profit**:

$$\max_{Q} \ \pi(Q) = p(Q) \cdot Q - C(Q).$$

- First-order condition (if interior solution): MR = MC.

Definition (Marginal Revenue, MR)

$\mathrm{MR}(Q) = \frac{d}{dQ}[p(Q) \cdot Q] = p(Q) + Q\, p'(Q).$

Monopoly Equilibrium

- **Monopolist sets** Q^* where MR = MC.
- Price p^* is then $p(Q^*) > MC(Q^*)$ (usually).
- Typically, $Q^* < Q_{\text{competitive}}$ and $p^* > p_{\text{competitive}}$.

Result

Monopoly **restricts output** to raise price and earn **positive economic profit**, if demand is sufficiently high.

Welfare Effects of Monopoly

- Consumer surplus is **lower** due to higher price.
- Producer surplus (monopoly profit) is **higher** relative to perfect competition—some consumer surplus is transferred to producer.
- **Deadweight loss (DWL)**: The reduction in total surplus from restricting output below the competitive level.

Policy Implication

Antitrust laws may try to limit or regulate monopolies to reduce DWL and protect consumers.

Examples of Monopoly

- **Local utilities** (electricity, water) often are natural monopolies (high fixed costs + scale economies).
- **Patent holders**: Pharmaceutical firms with exclusive rights often set price above marginal cost.
- **Tech platforms** with network effects can exhibit monopoly-like dominance (though more complex).

Regulation or Public Ownership

Can be used when a single firm is more efficient (natural monopoly) to prevent excessive pricing.

Price Discrimination: Basic Idea

- A monopolist (or any firm with market power) charges **different prices** to different customers/groups.
- Goal: Extract more consumer surplus, increase profit.

Condition for Price Discrimination

- Firm must have market power. - Must be able to **segment** consumers (based on elasticity, willingness-to-pay, etc.). - Limited resale possibility (or it undermines the strategy).

Price Discrimination: First-Degree

First-Degree (Perfect) Price Discrimination:

- Firm charges each consumer their **maximum willingness to pay** for each unit.
- In theory, the monopolist captures **all** consumer surplus.
- Typically requires detailed information about each consumer's valuation.

Example (Example)

A car dealer attempting to negotiate individually with each buyer, extracting the maximum each is willing to pay. (Hard to achieve perfectly in practice.)

Price Discrimination: Second-Degree

Second-Degree (Nonlinear) Price Discrimination:

- Prices vary with the **quantity purchased** or in product "versioning" (quality differentiation).
- Firm does **not** observe each consumer's type but designs a menu of pricing schemes.

Example (Examples)

- *Quantity discounts: "Buy 2, get 1 half off."*
- *Different data plans in telecom: same product but "unlimited plan" vs. "basic plan."*

Price Discrimination: Third-Degree

Third-Degree (Group) Price Discrimination:

- The market is separated into segments with different demand elasticities.
- Monopolist charges a higher price to the group with the **less elastic** demand.
- Common in many consumer-oriented industries.

Example (Examples)

- *Student/senior discounts vs. regular adult pricing.*
- *Geographic price variations (e.g., textbooks in developing countries).*

Price Discrimination: Welfare Aspects

- **First-degree** can be *efficient* (no DWL) but transfers all surplus to the firm.
- **Second-degree** or **third-degree** can improve total output compared to uniform monopoly price, but typically some consumer surplus is transferred to producer.
- **Policy Perspective**: The effect on social welfare depends on whether increased output offset consumers' surplus loss.

Real-World Takeaway

Price discrimination is widespread (airline tickets, coupons, bundling). Regulators evaluate if it increases or reduces welfare, and whether it's "fair."

Oligopoly: A Few Major Players

- Market is dominated by a small number of firms, each with significant share.
- They must consider rivals' actions—**strategic interaction**.
- Various models: Cournot, Bertrand, Stackelberg, etc.

Link to Game Theory

Game-theoretic tools are crucial in analyzing oligopolistic competition.

Cournot Model: Quantity Competition

- Firms choose **quantities** $q_1, q_2, \ldots, q_n$ **simultaneously**.
- Inverse demand: $p = p(Q)$ where $Q = \sum_{i=1}^{n} q_i$.
- Each firm i solves

$$\max_{q_i \geq 0} \ \pi_i = p\big(Q\big)\, q_i - C_i(q_i),$$

taking q_{-i} (the rivals' outputs) as given.

Nash Equilibrium

Each q_i^* satisfies the best-response condition to q_{-i}^*. Typically found by solving a system of best-response equations.

Cournot Model: Insights

- **Equilibrium Price** is above marginal cost (unless n is large).
- Total output Q^* is *between* monopoly level and perfect competition level.
- As n grows, competition intensifies, pushing the equilibrium closer to the competitive outcome.

Example (Basic 2-Firm Example)

- *Inverse demand:* $p(Q) = a - Q$.
- *Costs:* $C_i(q_i) = c\, q_i$ *(constant MC).*
- *Solve each firm's best-response function* $q_i^* = \frac{a-c-q_{-i}}{2}$.

Then $q_1^* = q_2^* = \frac{a-c}{3}$, $Q^* = \frac{2(a-c)}{3}$, *and* $p^* = a - \frac{2(a-c)}{3}$.

Bertrand Model: Price Competition

- Firms choose **prices** $p_1, p_2, \ldots, p_n$ simultaneously for a (largely) homogeneous product.
- Consumers buy from the firm with the **lowest price**.
- If prices are identical, demand is split (various tie-breaking rules).

Nash Equilibrium (Homogeneous Product)

If all firms have the same marginal cost c, the unique equilibrium price is $p^* = c$. Even with two firms, price can be driven down to marginal cost—resembling perfect competition.

Bertrand Insights and Extensions

- **Key Intuition**: Undercutting your rival's price (by a small amount) captures the entire market. So price competition is "fierce."
- **Extensions**:
  - *Capacity constraints* can soften competition (if a firm cannot serve the whole market).
  - *Product differentiation* also reduces the incentive to undercut drastically.
- **Welfare**: Price is driven close to MC, benefiting consumers relative to Cournot (though real markets may differ).

Stackelberg Model: Sequential Quantity Setting

- Two-stage game:
  1. **Leader** firm chooses its quantity first.
  2. **Follower** firm observes leader's choice and then picks its quantity.
- The follower solves a Cournot-like problem given q_{leader}, but the leader **anticipates** the follower's best response.

Leader's Advantage

Leader can commit to a certain output level, effectively forcing the follower to accommodate. Often yields higher profit for the leader than in Cournot.

Stackelberg Equilibrium Example (2 Firms) — Detailed

Setup:

- Inverse demand: $p(Q) = a - Q$.
- Cost: $C_i(q_i) = c\, q_i$ (constant marginal cost).
- **Stage 2 (Follower)**: Given q_L, follower solves a Cournot-like problem:

$$\max_{q_F \geq 0} \; \pi_F = \left[a - q_L - q_F\right] q_F - c\, q_F.$$

- Follower's best response:

$$q_F^*(q_L) = \frac{a - c - q_L}{2}.$$

Stage 1 (Leader): Anticipates $q_F^*(q_L)$ and chooses q_L to maximize

$$\pi_L(q_L) = \left[a - q_L - q_F^*(q_L)\right] q_L - c\, q_L.$$

Stackelberg Equilibrium Example (2 Firms) — Detailed (cont.)

Substitute the follower's best response:

$$q_F^*(q_L) = \frac{a - c - q_L}{2}.$$

Thus the leader's profit function becomes:

$$\pi_L(q_L) = \left[a - q_L - \frac{a - c - q_L}{2}\right] \cdot q_L \; - \; c\, q_L.$$

$$= \left[\frac{a + c - q_L}{2}\right] q_L \; - \; c\, q_L \; = \; \frac{q_L(a + c - q_L)}{2} \; - \; c\, q_L.$$

$$= \frac{q_L(a - c) - q_L^2}{2}.$$

First-order condition:

$$\frac{d\pi_L}{dq_L} = \frac{(a - c) - 2q_L}{2} = 0 \quad \Longrightarrow \quad q_L^* = \frac{a - c}{2}.$$

Follower's quantity:

$$a - c - q_L^* \qquad a - c - \frac{a-c}{2} \qquad \frac{a-c}{2} \qquad a - c$$

Stackelberg Equilibrium: Interpretation

- **Leader's output:** $q_L^* = \frac{a-c}{2}$.
- **Follower's output:** $q_F^* = \frac{a-c}{4}$.
- **Total output:** $Q^* = q_L^* + q_F^* = \frac{3(a-c)}{4}$.
- **Resulting price:** $p^* = a - Q^* = a - \frac{3(a-c)}{4} = c + \frac{a-c}{4}$.

Comparison

- **Leader** produces *more* than in Cournot (where each firm produces $\frac{a-c}{3}$). - **Follower** produces *less* than in Cournot. - Total output is higher than the Cournot level $\left(\frac{2(a-c)}{3}\right)$ but still below perfect competition.

Summary of Oligopoly Models

- **Cournot (Quantity)**: Equilibrium output is between monopoly and perfect competition.
- **Bertrand (Price)**: If goods are identical and no capacity constraints, $p^* = MC$.
- **Stackelberg (Sequential)**: Leader can secure more profit; total output often higher than Cournot.

Takeaway

Each model highlights different strategic considerations. Real markets may combine elements: product differentiation, capacity limits, repeated interaction, etc.

Welfare Comparison: Competition vs. Monopoly vs. Oligopoly

- **Perfect Competition**: No DWL, $p = MC$, high consumer surplus.
- **Monopoly**: DWL arises, consumer surplus reduced, producer surplus increased.
- **Oligopoly**: Typically in-between; the more aggressively firms compete, the closer to perfect competition.

Policy Implications

- Competition policy fosters rivalry or regulates market power. - Oligopolies are monitored for collusion.

Real-World Oligopoly Cases

- **Airline routes**: A few carriers often dominate, leading to potential price wars or collusion.
- **Telecom market**: Spectrum auctions, capacity constraints, repeated games.
- **Automobile industry**: Capacity, brand loyalty, alliances, strategic investment.

Lesson

Understanding strategic interaction is crucial for both policy-making and firm strategies.

Recap of Key Concepts

- **Supply and Demand**: Partial equilibrium foundations.
- **Perfect Competition**: Benchmark with price-taking, zero long-run profit, efficiency.
- **Monopoly**: Single-firm market power $\rightarrow$ restricted output, higher price, potential DWL.
- **Price Discrimination**: First-, second-, and third-degree approaches.
- **Oligopoly Models**: Cournot, Bertrand, Stackelberg $\rightarrow$ strategic interaction among few firms.

Exercises (1)

1. **Supply-Demand Shift**: Suppose the demand function is $D(p) = 120 - 2p$ and supply is $S(p) = 30 + p$.
   - Find the equilibrium price and quantity.
   - If a \$5 per-unit tax is imposed on producers, how does this affect equilibrium?
2. **Perfect Competition**:
   - Consider a competitive firm with cost function $C(q) = 50 + 2q + 0.5q^2$. Find its short-run supply curve and explain why it is MC above AVC.
   - If market demand is $Q^D = 100 - 2p$ and there are 10 identical firms, solve for short-run equilibrium price/quantity.

Exercises (2)

1. **Monopoly Pricing**:
   - Demand: $p(Q) = 60 - Q$, Cost: $C(Q) = 10 + 5Q$.
   - Find the monopoly's profit-maximizing Q^* and p^*.
   - Calculate consumer surplus, producer surplus, and deadweight loss.
2. **Oligopoly Intro (Cournot)**:
   - Two firms have identical cost $C(q) = 4q$, and face inverse demand $p = 50 - Q$.
   - Derive each firm's best-response function. Solve for Cournot equilibrium output (q_1^*, q_2^*) and price.

Exercises (3)

1. **Stackelberg**:
   - In the same setup as the Cournot example, assume Firm 1 moves first. Derive the Stackelberg equilibrium and compare each firm's output and profit to the Cournot result.
2. **Price Discrimination**:
   - Give a real-life example of third-degree PD and discuss how the firm separates different consumer groups.
   - Explain why resale or "arbitrage" could undermine certain forms of PD.

References (1/2)

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Additional Notes

- This chapter's **foundation**—supply, demand, partial equilibrium—underlies real-world policy analysis.
- **Monopoly and oligopoly** models connect with deeper game-theoretic topics, auctions, and mechanism design.
- Real markets often have **incomplete information** or are dynamic—leading us to further chapters.

Key Takeaway

Knowing how markets set prices under varying conditions (competition vs. market power) is crucial for economic intuition and policy.

Concluding Remarks

- Competitive markets allocate resources efficiently (under ideal assumptions).
- Market power leads to inefficiencies (monopoly, oligopoly).
- Government policies and institutions can mitigate or sometimes create market distortions.

Next Topics

We will soon dive deeper into **strategic interaction** and **information economics**, building on these foundations.

Q & A

Any Questions?

Thank you for your attention!